

★ WEEK 7 · PART 2 of 3

Testing for Cointegration

From Eyeballing the Spread to a Formal Verdict

Engle-Granger two-step · the Johansen system test · the cointegration rank

A spread that looks stationary might just be lucky. Part 2 makes the judgment rigorous: two formal tests that decide whether a genuine long-run equilibrium really exists.

2

PART 2 · TESTING

Is It Really Cointegrated?

The Engle-Granger two-step residual test, the Johansen system approach and its cointegration rank, when to use each, and the practical pitfalls of testing.

From a Picture to a Test

Part 1 gave the idea; Part 2 makes it rigorous

1

We can now recognize cointegration

Two $I(1)$ series whose combination is stationary share a long-run equilibrium. The spread is our anchor — but a plot alone is not proof.

2

Eyeballing is not enough

A spread can look mean-reverting by chance over a short sample. We need a formal test with proper critical values.

3

Two standard tests do the job

Engle-Granger for a pair, Johansen for a system. Both ask the same question — is there a stationary combination? — with statistical rigor.

Part 2 in one line: prove the equilibrium is real before you trade it.

Prerequisite: Confirm Each Series Is $I(1)$

Cointegration is a statement about $I(1)$ series

Before testing for cointegration, verify that each series is individually $I(1)$ — non-stationary in levels, stationary in differences. Cointegration only makes sense for such series.

The check

- Run a unit-root test (ADF/KPSS) on each level — expect non-stationary.
- Run it on each difference — expect stationary.
- Both series should be $I(1)$ before you proceed.

Why it matters

- If a series is already stationary, cointegration is the wrong tool.
- If a series is $I(2)$, the setup changes entirely.
- Getting the integration order right is step zero.

The Testing Landscape

Two classic ways to test for cointegration

Engle-Granger

- A two-step, single-equation procedure.
- Regress one series on the other, then test the residual.
- Simple and intuitive — ideal for a single pair.

Johansen

- A system approach based on a vector error-correction model.
- Handles many series and finds how MANY equilibria exist.
- More powerful and symmetric, but more involved.

Step 1: Estimate the Long-Run Relationship

A simple regression of one level on the other

Step 1: $Y_t = \alpha + \beta X_t + u_t$ (OLS estimate of the long-run relationship)

What you do

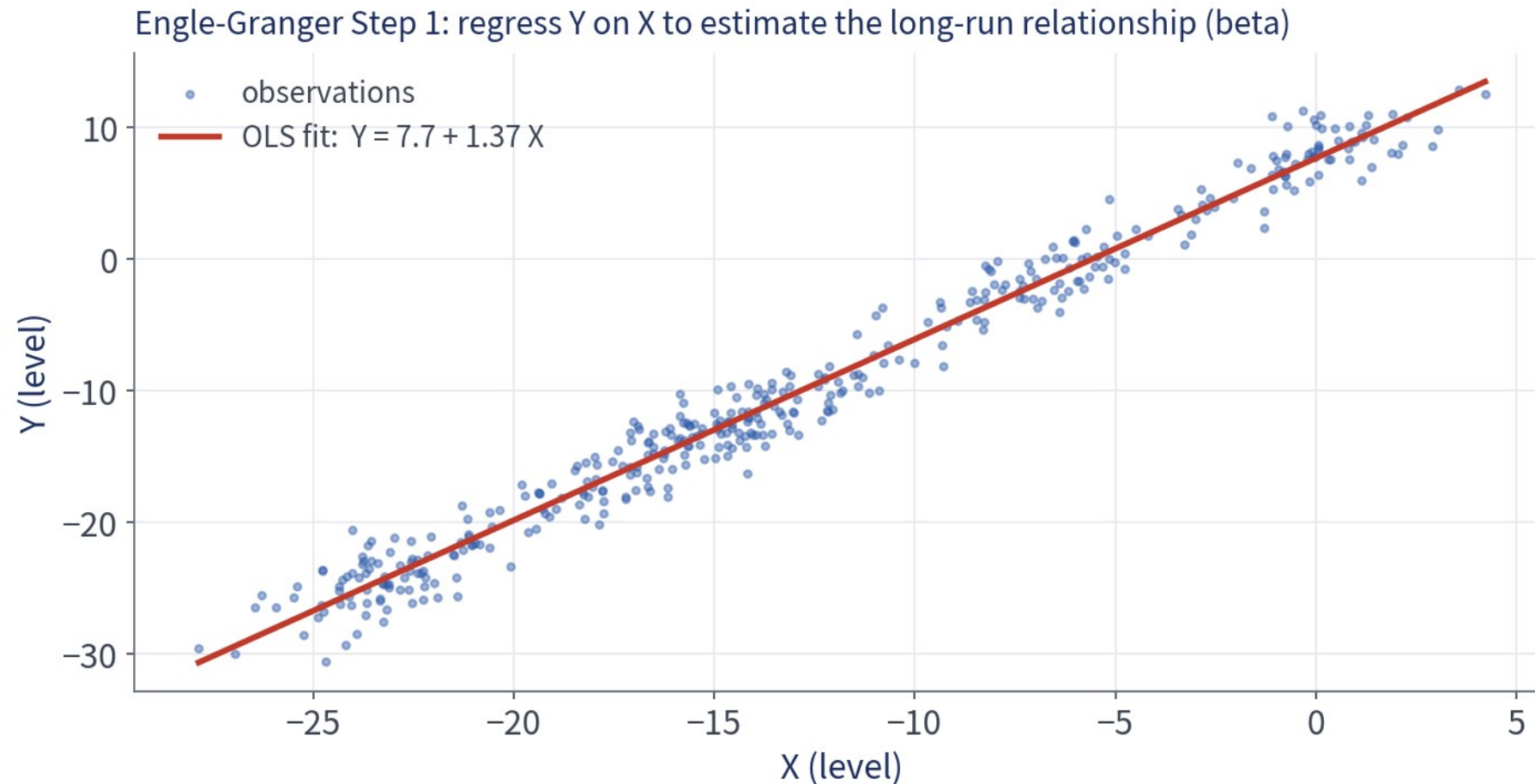
- Regress Y on X by ordinary least squares.
- The slope estimates the cointegrating coefficient β .
- The residual is the estimated equilibrium error.

What it gives

- A candidate long-run relationship between the levels.
- The spread $z = Y - \beta X$ to be tested next.
- This β is also the hedge ratio for trading (Part 3).

Step 1, Visualized

The regression line through the two levels estimates the long-run relationship. Its slope is β — the weight that (if the pair is cointegrated) makes the spread stationary.



Step 2: Test the Residual for Stationarity

The residual is the verdict

Step 2: ADF test on $\hat{u}_t$: H_0 : $\hat{u}_t$ has a unit root (NOT cointegrated)

What you do

- Run a unit-root (ADF) test on the step-1 residual.
- The null is that the residual has a unit root — NOT cointegrated.
- Reject the null $\rightarrow$ the residual is stationary $\rightarrow$ cointegrated.

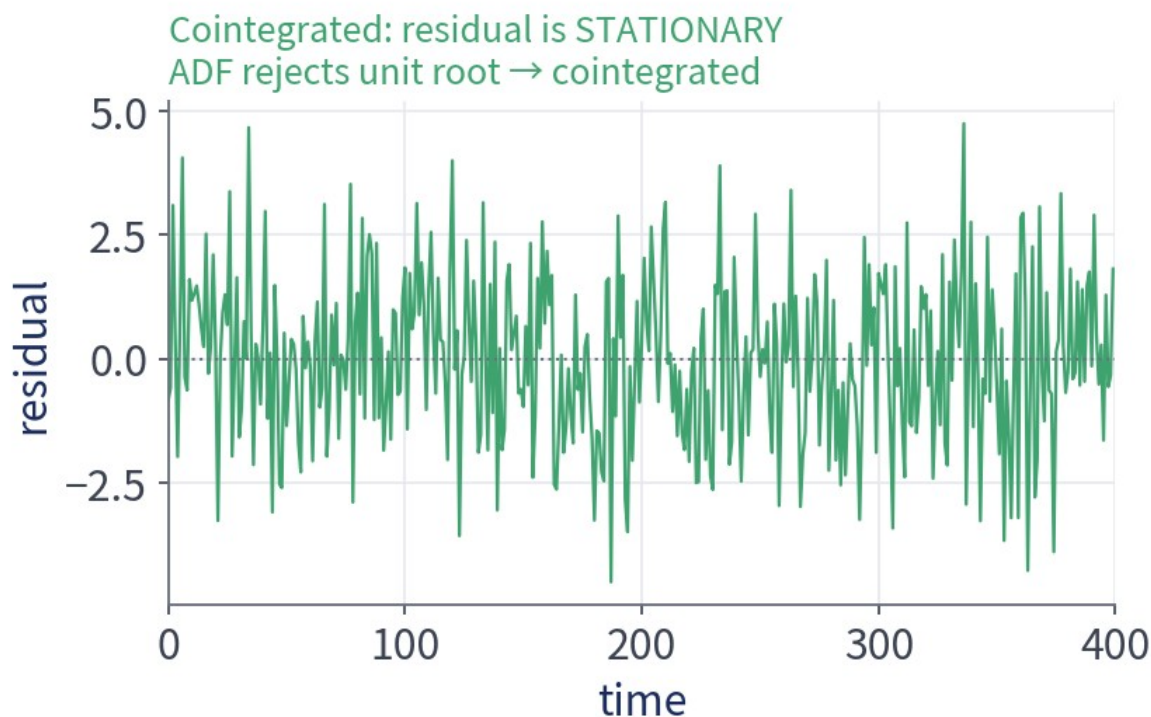
Why this is the whole test

- A stationary residual is the certificate of a real relationship.
- A wandering residual means the regression was spurious.
- Everything hinges on this one stationarity test.

Step 2, Visualized

The residual decides. For a cointegrated pair it mean-reverts and ADF rejects a unit root (left); for an unrelated pair it wanders and ADF cannot reject (right).

Engle-Granger Step 2: run an ADF test on the residual



Why Special Critical Values

The residual is estimated, not observed

The step-2 residual comes from a fitted regression, so ordinary ADF critical values don't apply. Engle-Granger uses its own, more demanding critical values.

The issue

- OLS chooses β to make the residual as small as possible.
- That biases it toward looking stationary.
- Standard ADF thresholds would over-reject the null.

The fix

- Use Engle-Granger (MacKinnon) critical values instead.
- They are stricter, correcting for the estimated β .
- Software applies the right ones automatically — just don't use plain ADF.

Which Series Is the Dependent One?

Y on X, or X on Y?

The problem

- Regressing Y on X and X on Y give different β estimates.
- In finite samples the test result can differ too.
- The method is not symmetric in the two variables.

How to handle it

- Try both orderings and check the conclusions agree.
- Prefer the more 'exogenous' series as the regressor if there is one.
- Or sidestep the issue entirely with Johansen.

The Limits of Engle-Granger

Great for a pair, awkward beyond it

Engle-Granger is simple and intuitive, but it can find only ONE relationship, works cleanly for just two series, and depends on which variable you place on the left.

What it can't do

- It finds at most one cointegrating relationship.
- With three or more series there may be several.
- It cannot tell you HOW MANY equilibria exist.

The remedy

- A system test that treats all series symmetrically.
- One that reports the NUMBER of relationships.
- That is exactly the Johansen approach.

The Johansen Test

A system approach for many series

Johansen fits a vector error-correction model directly and asks how many independent stationary combinations exist among the series — treating them all symmetrically.

What it does

- Works from the VECM, not a single regression.
- Estimates the number of cointegrating relations at once.
- Returns each cointegrating vector, not just one.

Why it is preferred for systems

- No arbitrary choice of dependent variable.
- It handles three or more series naturally.
- It is generally more powerful than Engle-Granger.

Where Johansen Looks

The rank of one matrix tells the whole story

$$\Delta \mathbf{Y}_t = \Pi \mathbf{Y}_{t-1} + \sum_i \Gamma_i \Delta \mathbf{Y}_{t-i} + \boldsymbol{\epsilon}_t, \quad \text{rank}(\Pi) = r$$

The key object

- Write the system as changes explained by past levels.
- The matrix on the lagged levels carries the equilibrium info.
- Its RANK equals the number of cointegrating relations.

The intuition

- If that matrix is zero, nothing ties the levels — no cointegration.
- If it has full rank, the series were stationary all along.
- In between: the rank counts the equilibria (Part 3 uses this).

The Cointegration Rank r

How many equilibria hold the system together?

$r = 0$: no cointegration $0 < r < k$: r relations $r = k$: all stationary

Reading r

- $r = 0$: no cointegration — nothing ties the series.
- $0 < r < k$: exactly r long-run relationships.
- $r = k$: the series were all stationary to begin with.

Why the count matters

- It tells you how many tradable equilibria exist.
- With k series you can have up to $k - 1$ relationships.
- Johansen finds this number; Engle-Granger cannot.

Determining the Rank

Step down until you can't reject

The procedure

- Test the null $r = 0$; if rejected, there is at least one relation.
- Then test $r \leq 1$, $r \leq 2$, and so on.
- Stop at the first null you cannot reject — that is the rank.

What you compare

- A test statistic against a critical value at each step.
- A statistic above the critical value rejects that null.
- The chain of rejections pins down r .

Trace vs Maximum-Eigenvalue

Two flavors of the Johansen test

The trace test

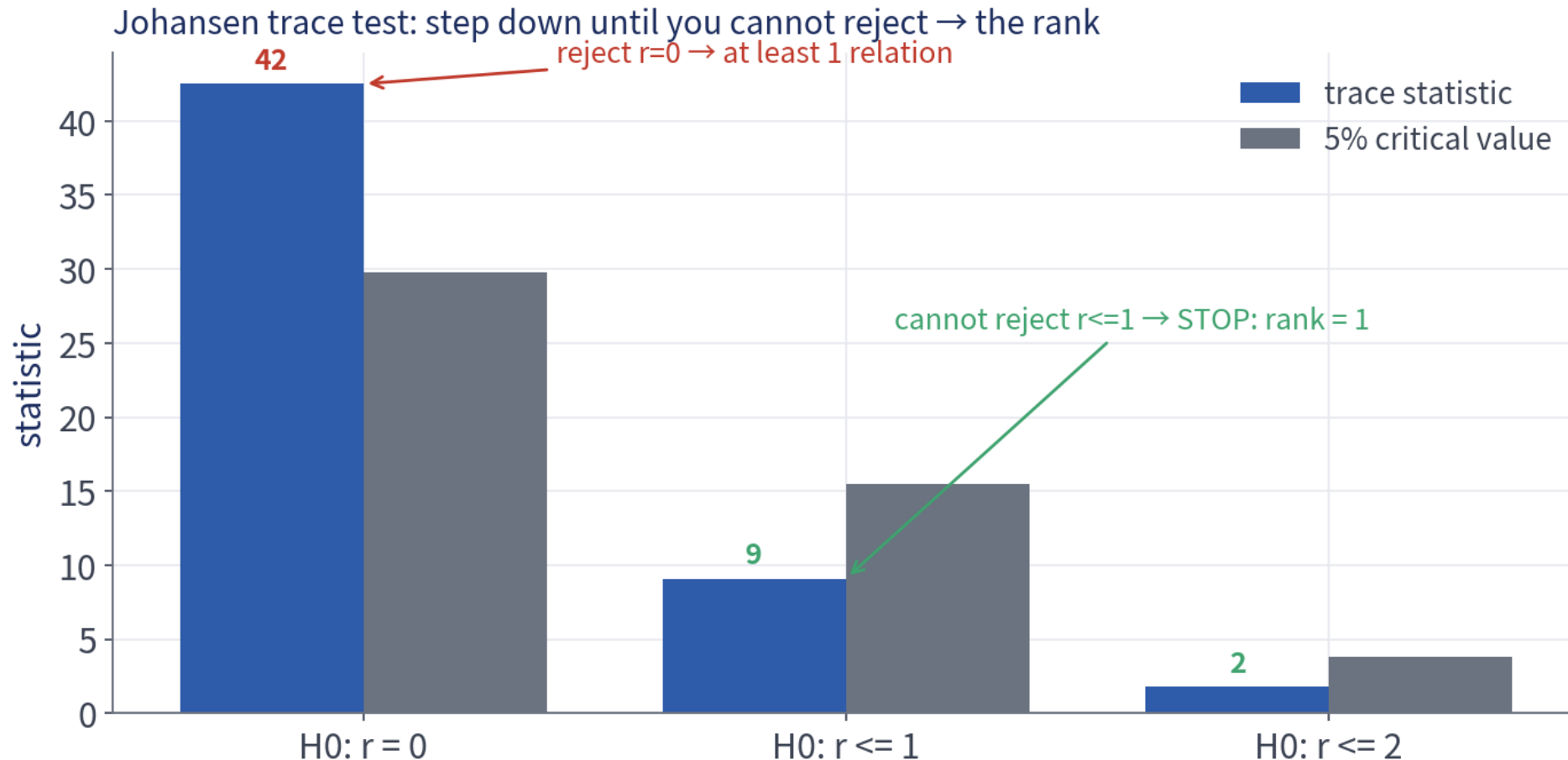
- Tests 'at most r relations' against 'more than r '.
- It pools information across the remaining directions.
- The most commonly reported version.

The max-eigenvalue test

- Tests 'exactly r ' against 'exactly $r + 1$ '.
- More focused on the single next relationship.
- The two usually agree; when they don't, investigate.

Reading the Trace Test

Reject $r = 0$ (statistic above the critical value) — so at least one relation exists. Fail to reject $r \leq 1$ — so stop: the cointegration rank is one.



Engle-Granger vs Johansen

Engle-Granger is a quick two-step check for a single pair; Johansen is a symmetric system test that finds how many equilibria exist. Match the test to the problem.

Engle-Granger vs Johansen

Pick the test that matches the number of series - and what you need from it.

Engle-Granger

a two-step, single-equation test

Best for

- exactly TWO series (one pair)
- a quick, intuitive check

How it works

1. regress Y on X (get beta)
2. ADF-test the residual

Limits

- finds at most ONE relationship
- result depends on which is Y
- less powerful in small samples

Johansen

a system (VECM-based) test

Best for

- THREE or more series
- finding several equilibria

How it works

- fits a VECM directly
- tests the rank of Π (trace / max-eig)

Strengths

- treats all series symmetrically
- finds the NUMBER of relations (rank)
- gives every cointegrating vector

Which Test, at a Glance

A one-look decision aid

Feature	Engle-Granger	Johansen
Number of series	2 (a pair)	2 or more
Approach	single-equation, 2-step	system (VECM-based)
Relations it finds	at most one	up to $k - 1$ (the rank)
Symmetric in variables	no (you pick Y)	yes

Use Engle-Granger for a fast pair check; use Johansen for many series or when the count of equilibria matters.

Practical Testing Pitfalls

Where cointegration tests mislead

Traps to watch

- Structural breaks can hide or fake cointegration.
- Short samples give low power — real links go undetected.
- The lag length in the test affects the result.

Good habits

- Confirm each series is $I(1)$ first, and inspect for breaks.
- Use enough data; be skeptical of borderline results.
- Choose lags by an information criterion, and check robustness.

The Cointegration Testing Workflow

Five steps to a defensible verdict

1 · Integration

each $I(1)$?

Unit-root test
each series;
confirm all are
 $I(1)$.

2 · Choose test

pair or system

Engle-Granger
for a pair;
Johansen for
more.

3 · Run it

proper values

Use the correct
(EG/Johansen)
critical values.

4 · Read result

rank / reject

Cointegrated?
How many
relations?

5 · Robustness

check

Vary lags,
orderings, sub-
samples; look
for breaks.

A positive result gives you β and the spread — the raw material for error correction and trading.

Test a Pair for Cointegration

Direct Claude Code through both tests, end to end

GOAL Run Engle-Granger and Johansen on a candidate pair, compare their verdicts, and read the rank.

[INPUT] Prompt · Claude Code

Write and run `coint_test.py` that:

1. confirms each series is $I(1)$ with ADF;
2. runs Engle-Granger BOTH ways ($Y \sim X$ and $X \sim Y$) using the proper critical values, reporting the p-values and estimated beta;
3. runs the Johansen test and prints the trace and max-eigenvalue statistics with the rank;
4. states the final verdict and the spread.

Note any disagreement between the two tests or between the two Engle-Granger orderings.

How to work with it

- Both series must be $I(1)$ for the test to be valid.
- EG and Johansen should broadly agree for a pair.
- The Johansen rank should be 0 or 1 for two series.
- Keep it; Part 3 trades the resulting spread.

Deliverable: `coint_test.py` + EG and Johansen results with a clear verdict.

Three Questions Before You Move On

If you can answer these, Part 2 has landed

Q1	What exactly does Engle-Granger's Step 2 test?	Residual stationarity
Q2	Johansen returns rank $r = 2$ for four series. What does that mean?	Two equilibria
Q3	Why can't you use ordinary ADF critical values in Step 2?	Residual is estimated

All three come from the two-step logic, the rank, and the estimated-residual subtlety.

Three Misconceptions to Avoid

Where testing goes wrong

Common mistakes

- "A stationary-looking spread proves cointegration."
- "Plain ADF critical values are fine for Step 2."
- "Engle-Granger and Johansen are interchangeable."

The reality

- Appearances deceive over short samples — run the formal test.
- Use Engle-Granger critical values; the residual is estimated.
- EG suits a pair; Johansen suits systems and counts the rank.

Three Things to Carry Forward

Part 2 in one screen

1

Engle-Granger is a two-step residual test

Regress one level on the other to get β , then test whether the residual is stationary — with special critical values, because the residual is estimated.

2

Johansen tests a whole system

It fits a VECM and reads off the cointegration rank — how many independent equilibria exist — treating all series symmetrically. Best for three or more.

3

Prove it before you trade it

Confirm each series is $I(1)$, pick the right test, use the right critical values, and check robustness. A positive result hands you β and the spread.

Next — Part 3: error correction, the VECM, the speed of adjustment, and pairs trading Stage II.